Lecture 6. Nonlinear Equations

Bisection, fixed-point iteration, contraction mapping, Newton's method

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We want x with $f(x) = 0$ for nonlinear f. There is no direct method; we iterate. Root finding and optimization are closely linked: a critical point of a differentiable g is a root of $f = g'$ (or ∇g). Statistical uses include maximum likelihood, nonlinear least squares, Bayes and MAP estimation. Examples: a root of $f(x) = x^4 - \sin x$; the maximizer of $g(x) = \log x/(1+x)$, a root of $g'(x) = \{1 + x^{-1} - \log x\}/(1+x)^2$.

6.1 Bisection

If f is continuous on $[l, r]$ with $f(l)f(r) \le 0$, the intermediate value theorem gives a root in $[l, r]$. Evaluate f at the midpoint $m = (l+r)/2$: stop if $f(m) = 0$; otherwise replace r by m if $f(l)f(m) < 0$ and l by m otherwise. After n steps the bracketing interval has length $2^{-n}(r^{(0)} - l^{(0)})$, so the midpoint is within that distance of a root. Bisection is slow (one bit per step) but always works under these weak assumptions. A typical use is computing a quantile: solve $F(x) - u = 0$ for a continuous cdf F. If F itself is only available as $\int_{-\infty}^{x} f$, each evaluation is a numerical integral (Lecture 8); this is how the inverse cdf method of Lecture 9 can be used when F^{-1} has no closed form.

6.2 Fixed-point iteration

x is a **fixed point** of F if $F(x) = x$. To find a root of f, choose F with $f(x) = 0 \iff F(x) = x$ and iterate $x^{(k+1)} = F(x^{(k)})$. (The iterative methods of Lecture 5 are the case $F(x) = Rx + c$.)

Theorem 6.1 (Contraction mapping)

Let $I \subset \mathbb{R}$ be a closed interval and $F : I \to I$ with $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$, where $0 \le L < 1$. Then F has a unique fixed point $x^\star \in I$, the iteration converges to it from every $x^{(0)} \in I$, and

$$|x^{(n)} - x^\star| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|.$$

Proof. $|x^{(k+1)} - x^{(k)}| = |F(x^{(k)}) - F(x^{(k-1)})| \le L|x^{(k)} - x^{(k-1)}| \le \cdots \le L^k|x^{(1)} - x^{(0)}|$. For $m > n$, by the triangle inequality,

$$|x^{(m)} - x^{(n)}| \le \sum_{k=n}^{m-1} L^k|x^{(1)} - x^{(0)}| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|. \qquad (*)$$

So the sequence is Cauchy and converges to some $x^\star$, which lies in I because I is closed. F is continuous, so $x^\star = \lim x^{(k+1)} = \lim F(x^{(k)}) = F(x^\star)$. If $y^\star$ were another fixed point, $|x^\star - y^\star| = |F(x^\star) - F(y^\star)| \le L|x^\star - y^\star|$ forces $x^\star = y^\star$. Letting $m \to \infty$ in $(*)$ gives the bound. ∎

The geometric rate L^n is called **linear convergence**.

Proposition 6.2

If F is differentiable on $I = [a, b]$, then $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$ if and only if $|F'(x)| \le L$ for all $x \in I$. Hence the best Lipschitz constant is $\sup_I |F'|$.

Proof. ($\Rightarrow$) For $x \in I$ and $y_n \to x$ in I, $y_n \ne x$, the difference quotients are bounded by L in absolute value, and so is their limit $F'(x)$.

($\Leftarrow$) For $x \ne y$, the mean value theorem gives c between them with $|F(x) - F(y)| = |F'(c)||x - y| \le L|x - y|$. ∎

Example 6.1 (Relaxed fixed-point iteration)

To solve $f(x) = 0$, iterate $x^{(t+1)} = x^{(t)} + \alpha f(x^{(t)})$, $\alpha \neq 0$, i.e. $F(x) = x + \alpha f(x)$. By Proposition 6.2 and Theorem 6.1, it converges whenever F maps an interval I around the root into itself and $\sup_I |1 + \alpha f'(x)| < 1$. This needs f' of constant sign on I; if, say, $0 < m \leq f' \leq M$ there, every $\alpha \in (-2/M, 0)$ works, and $\alpha = -2/(m + M)$ gives the best constant $L = (M - m)/(M + m)$. So the free parameter α turns f into a contraction that the plain iteration $x + f(x)$ need not be, and tunes the rate. Newton's method is the adaptive choice $\alpha = -1/f'(x^{(t)})$, which makes $F'(x^\star) = 0$. If $f = g'$ for a twice differentiable g to be minimized, the iteration with $\alpha = -\gamma$ is exactly gradient descent $x^{(t+1)} = x^{(t)} - \gamma g'(x^{(t)})$ (Lecture 7), and the condition above is the familiar step-size bound $\gamma < 2/\sup g''$.

Example 6.2

For $f(x) = x^4 - \sin x$ on $(0, 1]$, both $F_1(x) = (\sin x)^{1/4}$ and $F_2(x) = \sin x / x^3$ have the root as a fixed point. On $I = [0.2, 1]$, F_1 maps I into itself and $F_1'(x) = \cos x / \{4(\sin x)^{3/4}\}$ is decreasing with $0 < F_1'(1) < F_1'(0.2) < 1$, so F_1 is a contraction and the iteration converges. F_2 maps $[0.9, 1]$ onto the larger $[0.841, 1.075]$ with $F_2'(1) < -1$: it expands, and the iteration diverges. The choice of F matters.

6.3 Newton's method

Motivation. Fixed-point iteration converges only linearly. Newton's method replaces f by its tangent line at each step, which doubles the number of correct digits near a simple root; the price is computing derivatives and the risk of divergence from a bad start.

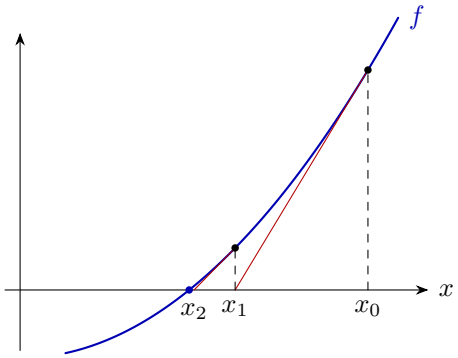


Figure. *Newton's method replaces f by its tangent line at x_k and takes the tangent's zero as $x_{k+1} = x_k - f(x_k)/f'(x_k)$. Near a simple root the error is roughly squared at each step.*

Linearizing f at the current iterate, $0 = f(x^\star) \approx f(x^{(k)}) + f'(x^{(k)})(x^\star - x^{(k)})$, suggests

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}.$$

This is fixed-point iteration with $F(x) = x - f(x)/f'(x)$, for which

$$F'(x) = \frac{f(x) f''(x)}{f'(x)^2}.$$

Theorem 6.3 (Local quadratic convergence)

Let $f(x^\star) = 0$, $f'(x^\star) \neq 0$, and f be three times continuously differentiable near $x^\star$. Then for $x^{(0)}$ close enough to $x^\star$, Newton's iterates converge to $x^\star$ and

$$\lim_{k\to\infty} \frac{|x^{(k+1)} - x^\star|}{|x^{(k)} - x^\star|^2} = \frac{1}{2}|F''(x^\star)| = \frac{|f''(x^\star)|}{2|f'(x^\star)|}.$$

Proof. $F'(x^\star) = 0$ since $f(x^\star) = 0$; by continuity $|F'| \le \frac{1}{2}$ on a closed neighbourhood I of $x^\star$, which F then maps into itself, so Theorem 6.1 gives convergence. By Taylor's theorem,

$$x^{(k+1)} - x^\star = F(x^{(k)}) - F(x^\star) = F'(x^\star)(x^{(k)} - x^\star) + \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2 = \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2,$$

with $\tilde{x}$ between $x^{(k)}$ and $x^\star$. Divide and let $k \to \infty$; $F''(x^\star) = f''(x^\star)/f'(x^\star)$ follows by differentiating F'. ∎

Quadratic convergence roughly doubles the number of correct digits per step. But it is only *local*: from a poor starting point Newton's method can wander off, converge to an unintended root, or diverge (e.g. seeking a root of $\cos x$ from different starting points). Safeguards (combining with bisection, line search) are used in practice.

6.4 Systems of equations

For $f : \mathbb{R}^n \to \mathbb{R}^n$ with Jacobian $Jf(x) = (\partial f_i/\partial x_j)$, Newton's method is

$$x^{(k+1)} = x^{(k)} - [Jf(x^{(k)})]^{-1} f(x^{(k)}),$$

implemented by solving the linear system $Jf(x^{(k)})\Delta x = -f(x^{(k)})$ (e.g. by LU) at each step, never by inverting. Cheaper variants replace the Jacobian:

- modified Newton: the fixed $J = Jf(x^{(0)})$ (factor once);
- Jacobi: $\operatorname{diag} Jf(x^{(k)})$;
- Gauss–Seidel: the lower triangle of $Jf(x^{(k)})$;
- SOR: $\frac{1}{\omega}\operatorname{diag} Jf(x^{(k)})$ plus the strict lower triangle.

These converge more slowly (typically linearly) but each step is much cheaper.

6.5 Selected exercises

Motivation. Theorem 6.3 is local and assumes a simple root. These exercises show a setting where Newton's method converges from every start, what happens at a multiple root, and a classical application in which the error is squared exactly.

Exercise 6.1 (Global convergence for convex increasing functions)

Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable and convex with $f' > 0$, and let $f(x^\star) = 0$.
(a) Show that from every $x^{(0)}$ the Newton iterates satisfy $x^{(k)} \ge x^\star$ for $k \ge 1$, decrease monotonically from $k = 1$ on, and converge to $x^\star$.
(b) Apply this to compute $\log a$ $(a > 0)$ without logarithms.

Solution. [Supplement]

(a)

Step 1. (Iterates overshoot to the right.) A differentiable convex function lies above its tangents: $f(y) \ge f(x) + f'(x)(y-x)$. With $x = x^{(k)}$ and $y = x^{(k+1)} = x^{(k)} - f(x^{(k)})/f'(x^{(k)})$, the right side is 0, so $f(x^{(k+1)}) \ge 0 = f(x^\star)$. As f is strictly increasing, $x^{(k+1)} \ge x^\star$ for every $k \ge 0$.

Step 2. (Monotonicity.) For $k \ge 1$, $f(x^{(k)}) \ge 0$ and $f'(x^{(k)}) > 0$, so $x^{(k+1)} = x^{(k)} - f(x^{(k)})/f'(x^{(k)}) \le x^{(k)}$.

Step 3. (Limit.) The sequence $(x^{(k)})_{k\ge1}$ is nonincreasing and bounded below by $x^\star$, so it converges to some $\ell \ge x^\star$. Letting $k \to \infty$ in the iteration and using the continuity of f and f' with $f'(\ell) > 0$, $\ell = \ell - f(\ell)/f'(\ell)$, so $f(\ell) = 0$, and $\ell = x^\star$ because f has only one root.

(b) $f(x) = e^x - a$ is convex with $f'(x) = e^x > 0$ and root $\log a$. Newton's method reads

$$x^{(k+1)} = x^{(k)} - \frac{e^{x^{(k)}} - a}{e^{x^{(k)}}} = x^{(k)} - 1 + a\,e^{-x^{(k)}},$$

which by (a) converges to $\log a$ from every starting point, and quadratically by Theorem 6.3 ($f'(\log a) = a \ne 0$). ■

Exercise 6.2 (Newton's method at a multiple root)

Let $f(x) = (x - x^\star)^m g(x)$ with an integer $m \ge 2$, g continuously differentiable near $x^\star$ and $g(x^\star) \ne 0$.
(a) Show that Newton's method converges to $x^\star$ from every point close enough to $x^\star$, but only linearly: $\lim_k (x^{(k+1)} - x^\star)/(x^{(k)} - x^\star) = (m-1)/m$.
(b) Show that the modified iteration $x^{(k+1)} = x^{(k)} - m f(x^{(k)})/f'(x^{(k)})$ satisfies $|x^{(k+1)} - x^\star| \le C|x^{(k)} - x^\star|^2$ near $x^\star$.

Solution. [Supplement] Write $e = x - x^\star$. Then $f'(x) = e^{m-1}\{m g(x) + e g'(x)\}$, and the bracket tends to $m g(x^\star) \ne 0$, so it is nonzero on some interval $|e| < \delta_0$; there $f'(x) \ne 0$ for $x \ne x^\star$ and

$$\frac{f(x)}{f'(x)} = \frac{e\,g(x)}{m g(x) + e g'(x)}.$$

(If an iterate equals $x^\star$, the iteration has stopped at the root.)

(a) The Newton map $F(x) = x - f(x)/f'(x)$ satisfies

$$F(x) - x^\star = e\Big(1 - \frac{g(x)}{m g(x) + e g'(x)}\Big) = e\,\varphi(e), \qquad \varphi(x) = \frac{(m-1)g(x) + e g'(x)}{m g(x) + e g'(x)}.$$

φ is continuous with $\varphi(x^\star) = (m-1)/m < 1$, so there are $\delta \in (0, \delta_0)$ and $q < 1$ with $|\varphi| \le q$ on $|e| < \delta$. Starting there, $|e_{k+1}| \le q|e_k|$, so the iterates stay in the interval and $e_k \to 0$; then $e_{k+1}/e_k = \varphi(x^{(k)}) \to (m-1)/m$.

(b) For $G(x) = x - m f(x)/f'(x)$,

$$G(x) - x^\star = e\Big(1 - \frac{m g(x)}{m g(x) + e g'(x)}\Big) = e^2\,\psi(e), \qquad \psi(x) = \frac{g'(x)}{m g(x) + e g'(x)},$$

and ψ is continuous, hence bounded by some C, on a closed interval around $x^\star$. So $|e_{k+1}| \le C e_k^2$, and for $C|e_0| < 1$ (and $|e_0|$ within that interval) the iteration converges quadratically. The multiplicity m must be known; otherwise one may apply Newton's method to f/f', which has a simple root at $x^\star$. ■

Exercise 6.3 (Division by multiplication)

Let $a > 0$ and apply Newton's method to $f(x) = 1/x - a$.
(a) Show that the iteration is $x^{(k+1)} = x^{(k)}(2 - ax^{(k)})$, which uses no division.
(b) With $r_k = 1 - ax^{(k)}$, show that $r_{k+1} = r_k^2$.
(c) Conclude that $x^{(k)} \to 1/a$ if and only if $0 < x^{(0)} < 2/a$, that the convergence is then quadratic, and describe what happens otherwise.

Solution. [Supplement]

(a) $f'(x) = -1/x^2$, so $x - f(x)/f'(x) = x + x^2(1/x - a) = 2x - ax^2$.

(b) $1 - ax(2 - ax) = 1 - 2ax + a^2x^2 = (1 - ax)^2$.

(c) By (b), $r_k = r_0^{2^k}$ and $x^{(k)} = (1 - r_k)/a$. If $|r_0| < 1$, i.e. $0 < x^{(0)} < 2/a$, then $r_k \to 0$ and $x^{(k)} \to 1/a$, with

$$\left|x^{(k+1)} - \frac{1}{a}\right| = \frac{r_k^2}{a} = a\left|x^{(k)} - \frac{1}{a}\right|^2,$$

quadratic convergence (the number of correct bits doubles exactly). If $|r_0| = 1$ ($x^{(0)} \in \{0, 2/a\}$), then $r_k = 1$ and $x^{(k)} = 0$ for $k \geq 1$. If $|r_0| > 1$, then $r_k \to +\infty$ and $x^{(k)} \to -\infty$. Hardware and software division routines use this iteration after a table lookup supplies a starting value with small $|r_0|$. ■